

- 5 (a) State one differences between *progressive* waves and *stationary* waves.

Progressive waves transfer energy but not stationary waves [B1]

Particle in a progressive wave have the same amplitude but particles in a stationary wave have different amplitude, ranging from a maximum at antinodes to minimum at nodes. (B1)

Adjacent particles in a progressive wave are out of phase but adjacent particles between two consecutive nodes in a stationary are in phase. (B1)

- (b) Define *transverse* waves and *longitudinal* waves.

Particles in a transverse wave oscillate/vibrate normal/perpendicular to the direction of energy transfer of wave [B1]

Particles in a longitudinal wave oscillate/vibrate parallel to the direction of energy transfer of wave [B1]

.....
 [1]

- (c) (i) Explain why it would not be possible to polarise sound waves.

Sound waves are longitudinal waves. [B1]

Hence, sound waves in a gas or liquid cannot be polarised because the medium oscillates only along the direction of wave propagation or energy transfer (i.e. a single line), and hence cannot be restricted to a plane normal to the direction of energy transfer. [B1]

..... [2]

- (ii) Unpolarised light of intensity I_0 is incident on two polarising filters P and Q, as shown in Fig. 5.1 below. The transmission axis of filter P is aligned vertically. The intensity of the unpolarised light is halved after passing P.

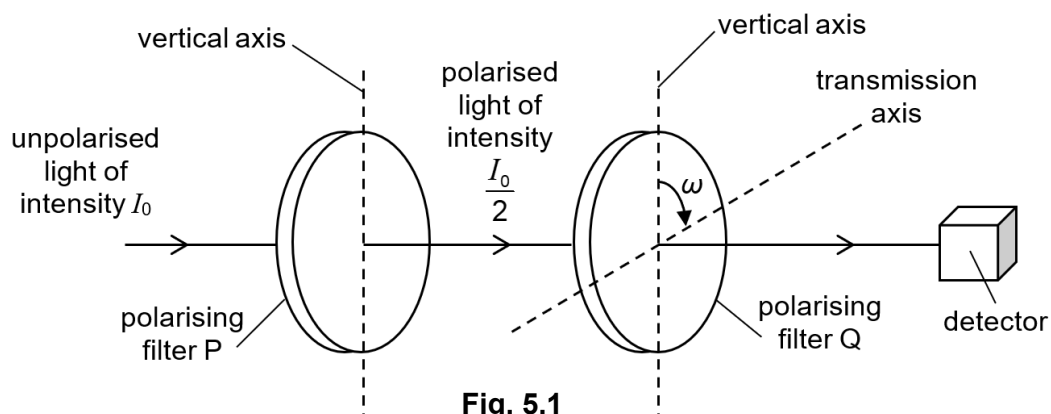


Fig. 5.1

The light is then passed through filter Q, which has the transmission axis initially aligned vertically and spun at a constant angular velocity of 2.0 rad s^{-1} .

Determine the ratio $\frac{\text{intensity reaching detector}}{\text{initial intensity } I_0}$ after 9.0 s.

relative angle between P and Q = $\omega t = 2(9) = 18 \text{ rad}$ [C1]

By Malus' law, $\frac{\frac{I_0}{2} \cos^2(18 \text{ rad})}{I_0}$ [B1]

= 0.218 [A1]

ratio = [3]

[Total: 8]

